

SSQ-LR: Sub-FP16 Recurrent State Quantization for Hybrid Reasoners

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Abstract

This is a living paper shell for a preregistered Mac-first experiment. SSQ-LR tests whether recurrent SSM state in hybrid reasoning models can be quantized below FP16 while preserving quality. No positive result or native systems claim is made yet; current held-out S2 scouts fail to produce a recipe that clears both quality and byte-accounting gates.

1 Current Gate and Claim Boundary

The branch must pass state-distribution heterogeneity, state-only quantization sensitivity, and cross-model transfer gates before any GPU validation. S1 must first show real state heterogeneity on a live hybrid model; S2 then freezes one state-only quantization recipe; S3 tests that recipe without per-model retuning. Failing any gate kills the branch before GPU work. The current real trace packet builder/checker covers S1. S2 and S3 now have executable follow-up contracts in `experimental/shared/followup_gate_contracts.py`: they require frozen recipe IDs, byte accounting, BF16 no-op controls, paired accuracy and continuation-NLL quality bounds, verbosity/length-drift readouts before revival, frozen recipe/source hashes, and no-retuning transfer rows. Resource-limited S2 scouts are not current evidence for promotion. SSQ-LR makes no quality, byte-savings, transfer, GPU, latency, or throughput claim until non-resource-limited S1–S3 packets clear.

2 Mac Artifact Status

A deterministic synthetic S1 packet now validates the real-schema checker path before real hybrid-model state dumps. It emits 288 rows covering 12 prompts, 6 recurrent layers, and the four preregistered buckets, producing `SCHEMA_REHEARSAL_NOT_PROMOTABLE_SYNTHETIC_SSQ_LR_S1`. This is synthetic-only: it is not model evidence, not a quality result, and not GPU evidence. The held-out S1b packet is the strongest positive evidence so far: it repeats the frozen selected layers 0, 12, and 30 on 12 held-out prompts with selected ratio 2.459 and lower bound 1.861. Uniform state recipes do not yet convert that heterogeneity into a method. MXFP4 preserves short BF16-argmax fidelity but reaches only $3.765\text{--}3.938\times$ state-memory reduction after scale bytes. INT3 reaches $4.923\text{--}5.224\times$ after scale bytes, but both 12-prompt held-out scouts fail because at least one prompt loses argmax fidelity. The 4-prompt INT3 pass is therefore treated as resource-limited and non-promoting. The newest resource-limited mixed-block S2b scout selects `mixed_int3_mxfp4_low_error_25pct`: it reaches $4.192\times$ counted state-memory reduction on the 12-prompt held-out replay with zero BF16-argmax delta and 0.03956 selected NLL-delta CI high. This short-window pass is still simulated replay and cannot promote to GPU. The same recipe then fails a longer-window replay (`--max-input-tokens 24 --prefix-tokens 8`): it preserves the $4.192\times$ byte readout but has selected accuracy CI high 0.0667 and selected NLL-delta CI high 0.0764. The mixed recipe is therefore stopped before GPU validation. Layer-localization gives one narrower layer set: layers 0 and 30 individually pass the same longer-window replay with INT3 state at $5.224\times$ counted memory reduction and zero selected accuracy delta, while layer 12 fails. The combined layers 0 and 30 replay also passes with selected NLL-delta CI high 0.04294. A stricter prefilter replay on this layer set weakens pure INT3 (accuracy CI high 0.105).

but selects `mixed_int3_mxfp4_low_error_25pct` at $4.192\times$ counted memory, zero selected accuracy drift, and 0.05044 selected NLL-delta CI high.

3 Next Evidence

The next admissible action is not GPU validation. SSQ-LR should freeze the layer-selective recipe, `mixed_int3_mxfp4_low_error_25pct` on layers 0 and 30 only, and test no-retuning transfer plus verbosity/length-drift readouts. The current S3 prefilter packet freezes this recipe and validates cleanly under the S3 schema, but fails the transfer gate because the local cache contains only one complete hybrid validation model. Granite Small, Granite Small FP8, and Qwen3-Next are config-only caches on this Mac. Otherwise the S1b/S2b result is diagnostic evidence, not a positive method.

4 Admissible Real Packet

Promotable packets must pin model/tokenizer revisions, prompt and trace-plan hashes, the shared architecture-map hash, exact command/device/dtype metadata, and saved recurrent-state tensor hashes. S1 rows must cover all four registered position buckets, including `prefill_end`, for every prompt/layer pair and at least 12 prompt IDs unless explicitly marked resource-limited. The checker recomputes row metrics and gate aggregates from raw tensors, rejects stale summaries, and requires `RESOURCE_LIMITED_NOT_PROMOTABLE` decisions for any subset packet. Full schema details live in the README and runbook.

Required baseline	Purpose
BF16 no-op replay	Catch hook/serialization bugs.
FP16 stochastic-rounding SSM cache	Deployment baseline from Nemotron-style hybrid serving.
INT16 block-scaled SSM cache	Higher-precision recurrent-cache control.
INT8 state-only	Standard low-bit state control.
FP8-style state-only	Checks whether sub-FP16 state can match the likely deployed fallback.
MXFP4 state-only	Candidate sub-FP16 recipe.
INT3 state-only	Tests whether sub-4-bit state can clear scale-byte overhead.
Random same-L2 noise	Separate structure from noise magnitude.
Scale shuffle	Test scale assignment sensitivity.
Byte accounting	Include scale/metadata overhead.

Table 1: Controls required before SSQ-LR can claim a state-quantization recipe.

5 Reviewer Packet

The reviewer-facing claim boundary is tracked in `experimental/ssq_lr/paper/reviewer_pack.md`. It records that SSQ-LR is not camera-ready as a method paper until real S1–S3 evidence exists. The newest novelty baseline is Nemotron 3 Super’s recurrent-cache recipe: NVIDIA documentation treats Mamba SSM cache quantization as a separate deployment challenge and selects FP16 with stochastic rounding for the SSM cache, with INT16/block-scaled variants discussed as controls. SSQ-LR is therefore not novel for noticing SSM cache precision; it must beat this baseline with a sub-FP16 recipe and paired quality evidence. The source memo is `../..../references/754_ssq_lr_nemotron_state_cache_refs_20260507.md`.

6 Claim Boundary

SSQ-LR is only about recurrent SSM-state precision in hybrid reasoners. It does not claim weight quantization, activation quantization, KV-cache quantization, native serving speed, or memory savings until a real state recipe and later GPU validation pass their preregistered gates.